what is diff in sort and sorted

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Did you mean: what is ***difference*** in sort and sorted

AI Overview தமிழ்

In Python, `sort()` and `sorted()` are both used to arrange data in a specific order, but they differ significantly in **how** they work, **what** they work on, and their **return values**.

The core difference is that `sort()` **sorts in-place** (modifying the original list), while `sorted()` **returns a new sorted list** (leaving the original untouched).

Key Differences at a Glance

Feature	`list.sort()` **(Method)**	`sorted()` **(Function)**
Object Type	Lists only (`list.sort()`)	Any iterable (list, tuple, dict, string, etc.)
Modification	**In-place** (modifies original)	**Original remains unchanged**
Return Value	`None`	A new sorted list

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Performance	Faster (no extra memory usage)	Slower (creates a copy)
Usage	`my_list.sort()`	`new_list = sorted(my_iterable)`

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Detailed Breakdown

1. `list.sort()` (In-Place)

- **Method of List:** It is a method specifically for lists.
- **Modifies Original:** It reorders the elements within the existing list and does not create a new one.
- **Returns `None`:** It returns `None`, which is a common pitfall for beginners who might try to assign it to a new variable.
- **Memory Efficient:** Because it sorts in-place, it is more efficient for large lists.

Example:

python

```python
numbers = [5, 2, 9, 1]
numbers.sort()
print(numbers)  # Output: [1, 2, 5, 9] (original list is changed)
```

2. `sorted()` (Function)

- **Built-in Function:** It is a built-in function that accepts any iterable (list, tuple, string, dictionary keys, etc.).
- **Creates New List:** It creates a new list from the elements of the original, leaving the original data as it was.
- **Returns List:** It always returns a new list, even if the input was a tuple or a string.
- **Versatile:** It is more flexible because it can handle any iterable.

Example:

python

```python
numbers = [5, 2, 9, 1]
new_list = sorted(numbers)
print(numbers)   # Output: [5, 2, 9, 1] (original list is unchange
print(new_list) # Output: [1, 2, 5, 9] (new sorted list)
```

When to Use Which?

- **Use** `.sort()` if you are working with a list, need maximum performance, and do not need to keep the original, unordered list.
- **Use** `sorted()` if you need to keep the original data intact, are working with a data structure other than a list (like a tuple or dictionary), or need to create a new, sorted variable.

Common Parameters (Both)

Both `sort()` and `sorted()` accept the same optional arguments to customize the sorting:

- `reverse=True`: Sorts the data in descending order.
- `key=func`: Accepts a function (often a `lambda`) to specify a custom sorting basis (e.g., `key=len` to sort by length).

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